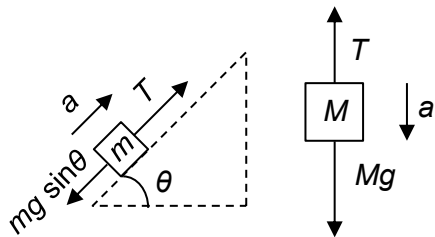


**Q4 Ans: D**



Let  $T$  be tension in string

$$T - mg \sin \theta = ma \dots\dots\dots(1)$$

$$Mg - T = Ma \dots\dots\dots(2)$$

$$(2) + (1)$$

$$Mg - mg \sin \theta = Ma + ma$$

$$a = \frac{Mg - mg \sin \theta}{M + m} = \frac{(M - m \sin \theta)}{(M + m)} g$$



